BIOPULSE PERSONA JUSTIFICATION

Persona Development Overview

Purpose

Personas were created to embody the main user groups recognized during the BioPulse requirements analysis stage. Instead of focusing on single users, typical personas were developed based on common objectives, motivations, behaviors, challenges, and technology usage trends.
The personas influenced the creation of user stories, user journeys, emergency workflows, accessibility factors, navigation frameworks, and high-fidelity screen designs during the BioPulse design workflow.

Persona Summary

Persona	User Group
Sarah Williams	Vulnerable Commuter
Michael Chen	Independent Traveller
Margaret Taylor	Independent Senior

Persona Profile

Attribute	Details
Name	Sarah Williams
Age	22
Occupation	University Student
Location	Auckland
Devices	iPhone, Apple Watch
Technology Confidence	High
Travel Frequency	Daily

Biography

Sarah is a third-year university student who regularly travels between home, university, and part-time work using public transport. She frequently commutes alone during early mornings and late evenings and often travels through unfamiliar areas. Although she is

comfortable using technology, she worries about personal safety during vulnerable situations where accessing her phone may be difficult.

User Needs

- Automatic emergency detection
- Fast emergency escalation
- Large emergency controls
- Concise emergency communication
- Reliable personal safety support

Goals

- Feel safe while travelling alone
- Receive emergency assistance quickly
- Minimise manual interaction during emergencies
- Notify emergency contacts automatically
- Receive clear emergency status updates

Pain Points

- Cannot access phone easily during emergencies
- Existing safety applications require manual activation
- Delayed emergency response may place users at greater risk
- Travelling alone at night increases safety concerns

Technology Proficiency

Technology	Proficiency
Smartphone Usage	90%
Wearables	80%
Mobile Apps	85%
Technical Knowledge	75%

Design Implications

- Large emergency controls
- Face ID disarm feature
- Automatic threat escalation
- High-visibility emergency alerts

Persona 02

Quote

"I want something running quietly in the background that only steps in when I actually need it."

Profile

Attribute	Details
Name	Michael Chen
Age	34
Occupation	Warehouse Supervisor
Location	Auckland
Devices	Android Smartphone
Technology Confidence	Moderate
Travel Frequency	Daily

Biography

Michael frequently finishes work late and travels home alone during the evening. While he considers himself independent, he appreciates technologies that provide additional protection without constant interaction. He values reliability, battery efficiency, and simplicity.

User Needs

- Background monitoring
- Reliable connection
- Simple onboarding
- Minimal maintenance

- Battery-efficient performance

Goals

- Passive safety monitoring
- Reliable wearable connectivity
- Long battery life
- Low maintenance
- Receive clear emergency status updates

Pain Points

- False alarms reduce trust in monitoring systems
- Complicated setup processes discourage adoption
- Constant notifications become distracting
- Unreliable wearable connectivity interrupts monitoring

Technology Proficiency

Technology	Proficiency
Smartphone Usage	80%
Wearables	60%
Mobile Apps	75%
Technical Knowledge	70%

Design Implications

- NFC quick pairing
- Background monitoring
- Battery-efficient workflows
- Minimal notification strategy

Persona 03

Quote

"I don't mind technology as long as it doesn't make me feel lost."

Profile

Attribute	Details
Name	Margaret Taylor
Age	67
Occupation	Retired
Location	Auckland
Devices	Android Smartphone
Technology Confidence	Basic
Travel Frequency	Weekly

Biography

Margaret regularly travels to medical appointments, shopping centres, and community events. She values independent living and wants technologies that increase her safety. However, she often struggles with complex interfaces, small text, and technical terminology.

User Needs

- Large typography
- High contrast colours
- Simple navigation
- Clear instructions
- Accessible interactions

Goals

- Use technology confidently
- Access support quickly
- Feel safe while travelling
- Easily understand emergency information
- Maintain independence

Pain Points

- Small text is difficult to read
- Technical terminology is confusing
- Complex interfaces are overwhelming
- Hidden functions reduce confidence when using applications

Technology Proficiency

Technology	Proficiency
Smartphone Usage	60%
Wearables	30%
Mobile Apps	40%
Technical Knowledge	35%

Design Implications

- High contrast colours
- Larger typography
- Simplified navigation
- Clear system feedback

Persona Mapping Summary

Representative User Mapping

Persona	Users Represented
Sarah Williams	Users 1–3
Michael Chen	Users 4–6
Margaret Taylor	Users 7–8

Why Personas Were Used

The eight representative users identified during the requirements analysis phase were grouped into three personas based on similar goals, behaviours, needs, and technology usage patterns.

This approach enabled the BioPulse design process to focus on broader user requirements while ensuring accessibility, emergency responsiveness, and usability considerations were incorporated throughout the system.

Design Influence

The personas directly informed:

- User Stories
- User Journey Maps
- Navigation Design
- Emergency Workflows
- Accessibility Requirements
- Screen Hierarchy
- Final High-Fidelity Prototypes

Conclusion

The personas established a user-centred foundation for the BioPulse design process. By understanding the needs of vulnerable commuters, independent travellers, and senior users, the system was able to prioritise personal safety, accessibility, emergency responsiveness, and ease of use. These personas informed all subsequent UX artefacts and design decisions throughout the project lifecycle.